



**SETEC INSTITUTE**

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**MANAGEMENT INFORMATION SYSTEM**

**GROUP: SV7**

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**Topic: Coffee Order System**

**Subject : SA II**

**MEMBER**

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**7.Hoerun Chhaileang**

**8.Heang Chheakheng**

**YEAR 3 SEMESTER 2**

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# I. System Requirement

## Role: Customer

### PAGE 1: Digital Menu ( Home )

Purpose: Discover items and browse the full menu selection from the table.

#### **FUNCTIONS:**

Include a search bar at the top to find specific items. Provide quick filters by category: "All", "Drinks", "Food", "Desserts", "Snacks" . Display a product grid with images, prices, and brief descriptions. Show "SOLD OUT" indicators on unavailable items. "Add to Cart" button ( + ) for each item.

PAGE 2: Your Order ( Cart ) Purpose: Review selected items and customize the order before paying.

#### **FUNCTIONS:**

Display a list of selected items with quantity adjustment ( +/- ). Include a Special Instructions field for allergies or preferences ( e.g., "No onions" ). Show a real-time Total Calculation. Provide payment method selection: "KHQR" or "Cash".

PAGE 3: Scan to PayPurpose: Complete the transaction using modern digital payment.

**FUNCTIONS:**

Display a dynamic QR code for Bakong/ABA/Wing apps.Show the exact amount to be paid in USD/KHR.Provide a payment simulation timer or status indicator.

PAGE 4: Order StatusPurpose: Track the progress of the order in real-time.

**FUNCTIONS:**

Display "Order Placed!" and "Payment Confirmed" messages.Show live status: "Waiting for kitchen to accept".Display a detailed summary of the order with Table ID and Payment status.

**Role: Admin**

PAGE 1: Sales AnalyticsPurpose: Monitor business performance and revenue trends.

**FUNCTIONS:**

Display total daily revenue and order count.S

how "Daily Revenue—Cash vs KHQR" bar chart. Display "Revenue Split" pie chart for payment methods.

List "Top Products" by sales volume.

PAGE 2: Menu & Category Management Purpose: Control the digital menu inventory and structure.

**FUNCTIONS:**

Add/Edit/Delete products with price and category.

Toggle product "Availability" ( Mark as Sold Out ). Create and manage product categories.

PAGE 3: System & Table Setup Purpose: Configure physical table mapping for QR ordering.

**FUNCTIONS:**

QR Link Generator: Input table number ( e.g., A5 ) to create a unique URL. Generate and export QR code images.

PAGE 4: Audit Log Purpose: Financial oversight and loss prevention.

**FUNCTIONS:**

List "At Risk" orders ( items marked as Served but Unpaid ).

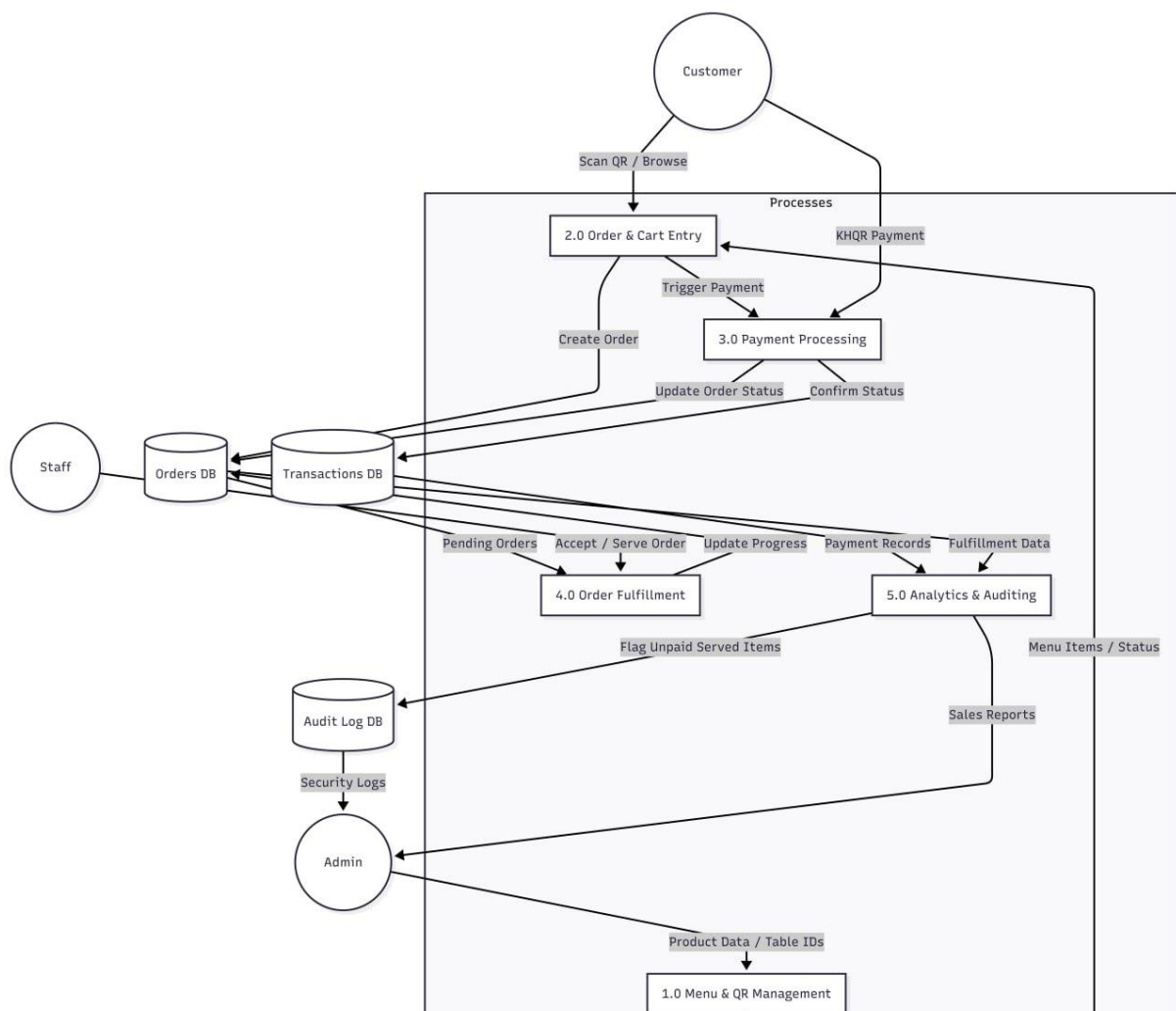
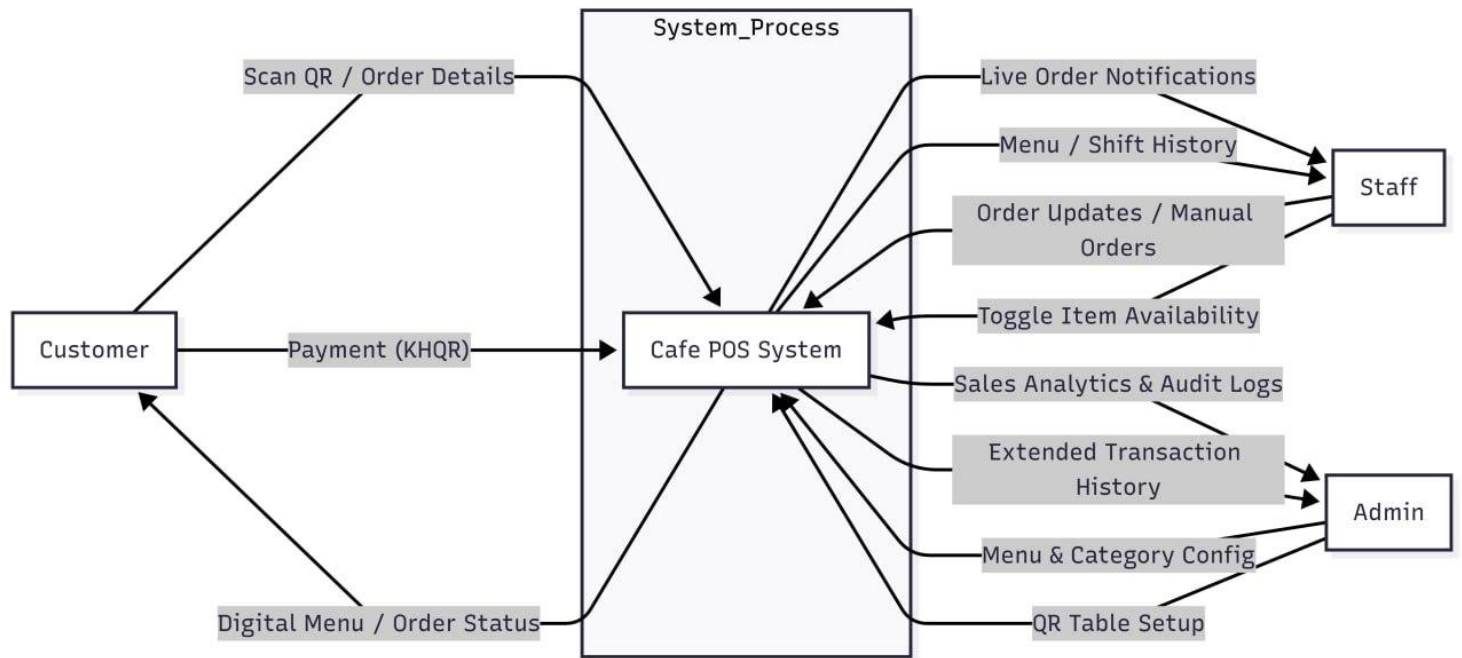
Display total amount at risk and associated Order ID/Staff. Filter logs by date or table ID.

PAGE 5: History ( All Transactions )Purpose: Provide a high-level audit trail of every order for financial reconciliation and performance review.

### **FUNCTIONS:**

Performance Summary:Display total order volume for the selected period ( e.g., "5 ORDERS" ).Show total "PAID REVENUE" currently collected.Search & Precision Filtering:Global search bar to locate specific transactions by Order ID ( e.g., ORD-005 ) or Table ( e.g., Table A5 )."All Status" filter to sort by payment or fulfillment progress.Date range selector to view historical data ( e.g., "Last 7 Days" or "Today" ).Transaction Log Table:Order Info: Display unique IDs, exact date/time stamps, and table location.Itemized List: Show quantity and name for every item in the order ( e.g., 2x Iced Coffee, 1x Crispy Spring Rolls ).Dual Status Tracking: View real-time status for both Payment ( PAID/UNPAID ) and Workflow ( NOT YET ACCEPTED, ACCEPTED, or COMPLETED ).Financial Breakdown: Display total amount per order and the specific payment channel ( via CASH or KHQR ).Document Management:"Print" button for each row to reprint lost or additional receipts.

## II. Data Flow Diagram

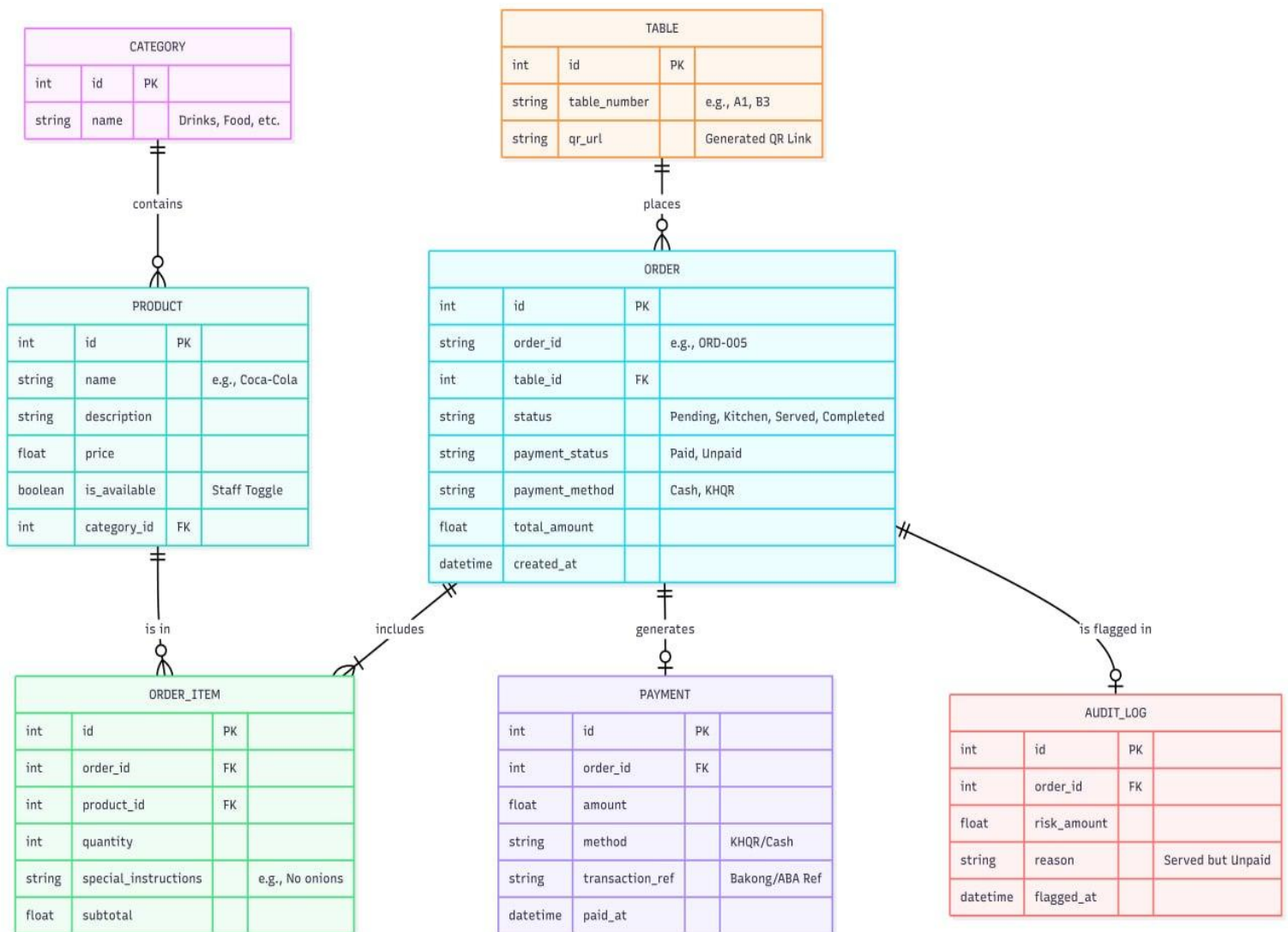


### III. Data Dictionary

| Entity     | Column       | Data Type | PK  | Nullable | Description                                                             |
|------------|--------------|-----------|-----|----------|-------------------------------------------------------------------------|
| CATEGORY   | id           | Integer   | Yes | No       | Unique ID for menu categories (e.g., Drinks, Food).                     |
|            | name         | String    | No  | No       | Category labels: "Drinks", "Food", "Desserts", "Snacks" .               |
| PRODUCT    | id           | Integer   | Yes | No       | Unique ID for each menu item.                                           |
|            | name         | String    | No  | No       | Name of the item (e.g., "Coca-Cola", "Iced Coffee").+1                  |
|            | description  | Text      | No  | Yes      | Product details (e.g., "Classic refreshing cola, served chilled").      |
|            | price        | Decimal   | No  | No       | Unit price in USD (e.g., 1.50, 6.50).+1                                 |
|            | is_available | Boolean   | No  | No       | Staff-controlled toggle to mark items as "Sold Out".                    |
|            | category_id  | Integer   | No  | No       | FK linking the product to its specific category.+1                      |
| TABLE      | id           | Integer   | Yes | No       | Unique ID for physical table locations.                                 |
|            | table_no     | String    | No  | No       | Table identifier used in the shop (e.g., "A1", "A5", "B3").+2           |
|            | qr_url       | String    | No  | No       | The unique URL generated for scanning at that specific table.           |
| ORDER      | id           | Integer   | Yes | No       | Unique system record ID.                                                |
|            | order_id     | String    | No  | No       | User-facing ID format: "ORD-001", "ORD-005".+1                          |
|            | table_id     | Integer   | No  | No       | FK linking the order to a physical table.+1                             |
|            | status       | Enum      | No  | No       | Workflow: "Pending", "In Kitchen", "Accepted", "Served", "Completed".+2 |
|            | pay_status   | Enum      | No  | No       | Financial state: "PAID" or "UNPAID".+1                                  |
|            | pay_method   | Enum      | No  | No       | Settlement type: "Cash" or "KHQR".+2                                    |
|            | total_amount | Decimal   | No  | No       | Total cost of the order.+1                                              |
|            | created_at   | DateTime  | No  | No       | Timestamp of when the order was placed.+1                               |
| ORDER_ITEM | id           | Integer   | Yes | No       | Unique ID for the specific line item.                                   |
|            | order_id     | Integer   | No  | No       | FK linking back to the parent order.                                    |
|            | product_id   | Integer   | No  | No       | FK linking to the specific product ordered.                             |
|            | quantity     | Integer   | No  | No       | Number of items (e.g., "1x", "2x", "3x").+1                             |
|            | instructions | Text      | No  | Yes      | Customer notes (e.g., "No onions", "Extra spicy").+2                    |
| PAYMENT    | id           | Integer   | Yes | No       | Unique transaction record ID.                                           |
|            | order_id     | Integer   | No  | No       | FK linking to the finalized order.                                      |
|            | amount       | Decimal   | No  | No       | Total amount paid by the customer.                                      |
|            | method       | Enum      | No  | No       | "Cash" or "KHQR".                                                       |
|            | trans_ref    | String    | No  | Yes      | Reference ID for KHQR/Bakong digital transactions .                     |
| AUDIT_LOG  | id           | Integer   | Yes | No       | Unique entry for security tracking.                                     |
|            | order_id     | Integer   | No  | No       | FK linking to the "At Risk" order.                                      |
|            | risk_amount  | Decimal   | No  | No       | Value of items marked "Served" but still "Unpaid".+1                    |
|            | reason       | String    | No  | No       | Flag description (e.g., "Orders marked as Served but unpaid").          |



## IV. Entity Relations Diagram



## V. Product Backlog

| ID                         | User Story                                                               | Estimation (h) | Priority  | Sprint | Status |
|----------------------------|--------------------------------------------------------------------------|----------------|-----------|--------|--------|
| Ordering Module (Customer) |                                                                          |                |           |        |        |
|                            | 1 As customer, I can scan a QR code to access the digital menu           |                | 8 High    |        | 1 Open |
|                            | 2 As customer, I can browse products by category (Drinks, Food, etc.)    |                | 12 High   |        | 1 Open |
|                            | 3 As customer, I can search for specific menu items                      |                | 8 Medium  |        | 1 Open |
|                            | 4 As customer, I can add items to my cart and adjust quantities          |                | 14 High   |        | 1 Open |
|                            | 5 As customer, I can add special instructions (allergies/preferences)    |                | 6 Medium  |        | 2 Open |
|                            | 6 As customer, I can choose a payment method (Cash or KHQR)              |                | 8 High    |        | 2 Open |
|                            | 7 As customer, I can pay via KHQR and view payment confirmation          |                | 16 High   |        | 2 Open |
|                            | 8 As customer, I can track my order status (Waiting/Accepted/Kitchen)    |                | 10 Medium |        | 3 Open |
| Staff Operations Module    |                                                                          |                |           |        |        |
|                            | 9 As staff, I can view a live dashboard of pending and in-kitchen orders |                | 20 High   |        | 2 Open |
|                            | 10 As staff, I can manually create an order for walk-in customers        |                | 14 Medium |        | 3 Open |
|                            | 11 As staff, I can update order status (Accept, Send to Kitchen, Served) |                | 12 High   |        | 2 Open |
|                            | 12 As staff, I can mark manual orders as "Paid" once cash is received    |                | 6 High    |        | 3 Open |
|                            | 13 As staff, I can print receipts for completed orders                   |                | 10 Medium |        | 4 Open |
| Admin & Management Module  |                                                                          |                |           |        |        |
|                            | 14 As admin, I can add, edit, or delete menu items and prices            |                | 16 High   |        | 3 Open |
|                            | 15 As admin, I can manage product categories and inventory counts        |                | 12 Medium |        | 4 Open |
|                            | 16 As admin, I can generate unique QR codes for specific table numbers   |                | 14 High   |        | 1 Open |
|                            | 17 As admin, I can view sales analytics (Revenue Split, Top Products)    |                | 20 Medium |        | 4 Open |
|                            | 18 As admin, I can view a history of all system transactions             |                | 10 Medium |        | 4 Open |
|                            | 19 As admin, I can access audit logs for unpaid "Served" orders          |                | 12 High   |        | 3 Open |

## VI. Sprint Backlog

| Id                                                   | Story Id | Description                                         | Response | Start Date | End | Estimation | Status |
|------------------------------------------------------|----------|-----------------------------------------------------|----------|------------|-----|------------|--------|
| Sprint 1                                             |          | Menu Access & Discovery                             |          |            |     |            |        |
| As admin, I can generate unique QR codes for tables  |          |                                                     |          |            |     |            |        |
|                                                      | 1        | 1 Design QR Generator UI (Angular)                  |          |            |     |            | 4 Open |
|                                                      | 2        | 1 Implement QR logic & Table Entity (Prisma/NestJS) |          |            |     |            | 6 Open |
|                                                      | 3        | 1 Create Unit Tests for QR generation               |          |            |     |            | 4 Open |
| As customer, I can scan a QR code to access the menu |          |                                                     |          |            |     |            |        |
|                                                      | 4        | 2 Create scanning landing page UI                   |          |            |     |            | 4 Open |
|                                                      | 5        | 2 Logic for URL table-ID detection                  |          |            |     |            | 4 Open |
| As customer, I can browse products by category       |          |                                                     |          |            |     |            |        |
|                                                      | 6        | 3 Create Menu Grid UI & Category Tabs               |          |            |     |            | 4 Open |
|                                                      | 7        | 3 Fetch products by category (NestJS API)           |          |            |     |            | 4 Open |
|                                                      | 8        | 3 Validation for "Sold Out" status                  |          |            |     |            | 4 Open |
| As customer, I can search for menu items             |          |                                                     |          |            |     |            |        |
|                                                      | 9        | 4 Add Search bar UI and filtering logic             |          |            |     |            | 4 Open |
|                                                      | 10       | 4 Create Automated UI test for search               |          |            |     |            | 4 Open |
| As customer, I can manage cart items                 |          |                                                     |          |            |     |            |        |
|                                                      | 11       | 5 Create Cart UI overlay                            |          |            |     |            | 4 Open |
|                                                      | 12       | 5 Logic for Add/Update/Remove items                 |          |            |     |            | 6 Open |
|                                                      | 13       | 5 Create Unit Test for cart totals                  |          |            |     |            | 4 Open |
| Sprint 2                                             |          | Ordering & Staff Dashboard                          |          |            |     |            |        |
| As customer, I can pay via KHQR                      |          |                                                     |          |            |     |            |        |
|                                                      | 14       | 6 Integrate Bakong/KHQR SDK/API                     |          |            |     |            | 8 Open |
|                                                      | 15       | 6 Design Payment Confirmation UI                    |          |            |     |            | 4 Open |
|                                                      | 16       | 6 Validation for payment success callback           |          |            |     |            | 4 Open |
| As staff, I can view live orders                     |          |                                                     |          |            |     |            |        |
|                                                      | 17       | 7 Create Live Dashboard UI (StaffDash)              |          |            |     |            | 8 Open |
|                                                      | 18       | 7 Implement WebSocket/Real-time updates             |          |            |     |            | 8 Open |
|                                                      | 19       | 7 Create Unit Tests for order fetching              |          |            |     |            | 4 Open |
| As customer, I can select payment method             |          |                                                     |          |            |     |            |        |
|                                                      | 20       | 8 Create Payment Selection UI (Cash/KHQR)           |          |            |     |            | 4 Open |
|                                                      | 21       | 8 Logic to map payment type to Order Entity         |          |            |     |            | 4 Open |
| As customer, I can add special instructions          |          |                                                     |          |            |     |            |        |
|                                                      | 22       | 9 Add Text Area for instructions in Cart            |          |            |     |            | 2 Open |
|                                                      | 23       | 9 Validation & Mapping instructions to DB           |          |            |     |            | 4 Open |
| As staff, I can update order status                  |          |                                                     |          |            |     |            |        |
|                                                      | 24       | 10 Logic for "Accept" and "In Kitchen" status       |          |            |     |            | 6 Open |
| Sprint 3                                             |          | Staff Operations & Audit                            |          |            |     |            |        |

|                                      |    |    |                                          |  |  |  |  |    |      |
|--------------------------------------|----|----|------------------------------------------|--|--|--|--|----|------|
|                                      | 31 | 14 | Logic to flag Unpaid vs. Served items    |  |  |  |  | 6  | Open |
| Payment Handling                     |    |    |                                          |  |  |  |  |    |      |
|                                      | 32 | 15 | Staff "Mark as Paid" logic for cash      |  |  |  |  | 6  | Open |
|                                      | 33 | 16 | Order tracking UI for customers (Part 1) |  |  |  |  | 2  | Open |
| Sprint 4                             |    |    | Analytics & Final Polish                 |  |  |  |  |    |      |
| As admin, I can view sales analytics |    |    |                                          |  |  |  |  |    |      |
|                                      | 34 | 17 | Design Dashboard with Revenue Charts     |  |  |  |  | 10 | Open |
|                                      | 35 | 17 | Logic for Top Products and Revenue Split |  |  |  |  | 10 | Open |
| As admin, I can manage categories    |    |    |                                          |  |  |  |  |    |      |
|                                      | 36 | 18 | CRUD UI for Categories (Food/Drinks)     |  |  |  |  | 6  | Open |
|                                      | 37 | 18 | Logic for inventory/stock counts         |  |  |  |  | 6  | Open |
| Reporting & Receipts                 |    |    |                                          |  |  |  |  |    |      |
|                                      | 38 | 19 | Implement Print Receipt functionality    |  |  |  |  | 10 | Open |
|                                      | 39 | 20 | Create Transaction History UI/Search     |  |  |  |  | 10 | Open |
|                                      | 40 | 16 | Order tracking UI for customers (Part 2) |  |  |  |  | 4  | Open |

## VII. User Interface

### Admin



Menu Management

13 items — manage availability & details

Search products...

All

Drinks

Food

Desserts

Snacks

| ITEM                                                                                 | CATEGORY | PRICE  | AVAILABLE   | ACTIONS                 |
|--------------------------------------------------------------------------------------|----------|--------|-------------|-------------------------|
| <div>Coca-Cola</div> <div>Classic refreshing cola, served chilled</div>              | DRINKS   | \$1.50 | <div></div> | <div></div> <div></div> |
| <div>Fresh Orange Juice</div> <div>Freshly squeezed, no preservatives</div>          | DRINKS   | \$3.00 | <div></div> | <div></div> <div></div> |
| <div>Iced Coffee</div> <div>Strong cold brew with milk</div>                         | DRINKS   | \$2.50 | <div></div> | <div></div> <div></div> |
| <div>Thai Milk Tea</div> <div>Creamy sweet tea with condensed milk</div>             | DRINKS   | \$2.75 | <div></div> | <div></div> <div></div> |
| <div>Sparkling Water</div> <div>Chilled sparkling mineral water</div>                | DRINKS   | \$1.00 | <div></div> | <div></div> <div></div> |
| <div>Beef Burger</div> <div>Angus beef patty with lettuce &amp; cheese</div>         | FOOD     | \$6.50 | <div></div> | <div></div> <div></div> |
| <div>Chicken Fried Rice</div> <div>Wok-fried rice with egg &amp; vegetables</div>    | FOOD     | \$4.50 | <div></div> | <div></div> <div></div> |
| <div>Grilled Salmon</div> <div>Fresh salmon fillet with steamed vegs</div>           | FOOD     | \$9.00 | <div></div> | <div></div> <div></div> |
| <div>Pad Thai Noodles</div> <div>Stir-fried rice noodles, shrimp &amp; pean...</div> | FOOD     | \$5.00 | <div></div> | <div></div> <div></div> |

+ Add Item

AD AdminDash

Dashboard

Products

Categories

Table Managements

History

Audit Log

Logout

Category Management

5 categories — used across 13 products

SYSTEM

All

0 products

CAT-1

Drinks

5 products

CAT-2

Food

4 products

CAT-3

Desserts

2 products

CAT-4

Snacks

2 products

CAT-5

Add Category

AD AdminDash

Dashboard

Products

Categories

Table Managements

History

Audit Log

Logout

System & Table Setup

QR Link Generator

Generator unique URLs for scanning :

Enter Table Number (e.g. A5)

Generator QR

AD AdminDash

Dashboard

Products

Categories

Table Managements

History

Audit Log

Logout

All Transactions

Admin view: Extended history and filtering

5

ORDERS

\$20.00

PAID REVENUE

Search by Order ID or Table...

All Status

Last 7 Days

| ORDER INFO                              | ITEMS                                                       | STATUS                     | TOTAL               | ACTIONS |
|-----------------------------------------|-------------------------------------------------------------|----------------------------|---------------------|---------|
| ORD-005<br>Mar 14, 11:03 AM<br>TABLE A5 | 1x Pad Thai Noodles<br>1x Mango Sticky Rice<br>3x Coca-Cola | UNPAID<br>NOT YET ACCEPTED | \$13.00<br>via CASH | Print   |
| ORD-001<br>Mar 14, 11:00 AM<br>TABLE A1 | 2x Coca-Cola<br>1x Beef Burger                              | UNPAID<br>NOT YET ACCEPTED | \$9.50<br>via CASH  | Print   |
| ORD-002<br>Mar 14, 10:53 AM<br>TABLE B3 | 2x Iced Coffee<br>1x Crispy Spring Rolls                    | PAID<br>ACCEPTED           | \$8.00<br>via KHQR  | Print   |
| ORD-003<br>Mar 14, 10:25 AM<br>TABLE A3 | 1x Grilled Salmon<br>1x Fresh Orange Juice                  | PAID<br>COMPLETED          | \$12.00<br>via KHQR | Print   |
| ORD-004<br>Mar 14, 10:05 AM<br>TABLE B2 | 2x Chicken Fried Rice<br>2x Thai Milk Tea                   | UNPAID<br>COMPLETED        | \$14.50<br>via CASH | Print   |

AD AdminDash

Dashboard

Products

Categories

Table Managements

History

Reports

Logout

Audit Log

Orders marked as Served but payment not collected

1 unpaid at risk: \$14.50

Action Required: These orders are completed but have not been paid. Total at risk: \$14.50. Review with staff immediately.

Search by order ID or table...

Date

| ORDER ID | TABLE    | ITEMS                                  | TOTAL   | PAYMENT       | TIME             | ACTION  |
|----------|----------|----------------------------------------|---------|---------------|------------------|---------|
| ORD-004  | TABLE B2 | Chicken Fried Rice x2 Thai Milk Tea x2 | \$14.50 | UNPAID - CASH | Mar 14, 09:29 AM | Receipt |

# Staff

ST StaffDash

Orders

Create Order

Products

History

ST StaffDash

Orders

Create Order

Products

History

Logout

Live Dashboard

Wednesday, March 18

5

TOTAL ORDERS

2

PENDING

1

IN KITCHEN

\$20

REVENUE (PAID)

Pending

2

ORD-005

Table A5

UNPAID

CASH

Waiting

Pad Thai Noodles

x1

\$5.00

Mango Sticky Rice

x1

\$3.50

Coca-Cola

x3

\$4.50

\$13.00

2m ago

Accept

Mark Paid

Receipt

ORD-001

Table A5

UNPAID

CASH

Waiting

Coca-Cola

x2

\$3.00

Beef Burger

x1

\$6.50

No orders please

\$9.50

5m ago

Accept

Mark Paid

Receipt

In Kitchen

1

ORD-002

Table B3

PAID

KHQR

In Kitchen

Iced Coffee

x2

\$5.00

Crispy Spring Rolls

x1

\$3.00

\$8.00

12m ago

Finish

Receipt

Served

2

ORD-003

Table A1

PAID

KHQR

Served

Grilled Salmon

x1

\$9.00

Fresh Orange Juice

x1

\$3.00

\$12.00

40m ago

Receipt

ORD-004

Table B2

UNPAID

CASH

Served

Chicken Fried Rice

x2

\$9.00

Thai Milk Tea

x2

\$5.50

Extra salty

\$14.50

1h 0m ago

Mark Paid

Receipt

Our Menu

CAFÉ POS

Table A1

Search Items...

All

Drinks

Food

Desserts

Snacks

Coca-Cola

Classic refreshing cola, served chilled

\$1.50

+

Fresh Orange Juice

Freshly squeezed, no preservatives

\$3.00

+

Iced Coffee

Strong cold brew with milk

\$2.50

+

Thai Milk Tea

Creamy sweet tea with condensed milk

\$2.75

+

SOLD OUT

Sparkling Water

Chilled sparkling mineral water

\$1.00

+

Beef Burger

Angus beef patty with lettuce & cheese

\$6.50

+

Chicken Fried Rice

Wok-fried rice with egg & vegetables

\$4.50

+

Grilled Salmon

Fresh salmon fillet with steamed vegs

\$9.00

+

Pad Thai Noodles

Stir-fried rice noodles, shrimp & peanuts

\$5.00

+

Mango Sticky Rice

Classic Thai dessert with sweet mango

\$3.50

+

SOLD OUT

Chocolate Lava Cake

Warm cake with gooey chocolate center

\$4.00

+

Crispy Spring Rolls

Fried golden rolls with sweet chili sauce

\$3.00

+

Loaded Fries

Seasoned fries with cheese sauce & jalapeños

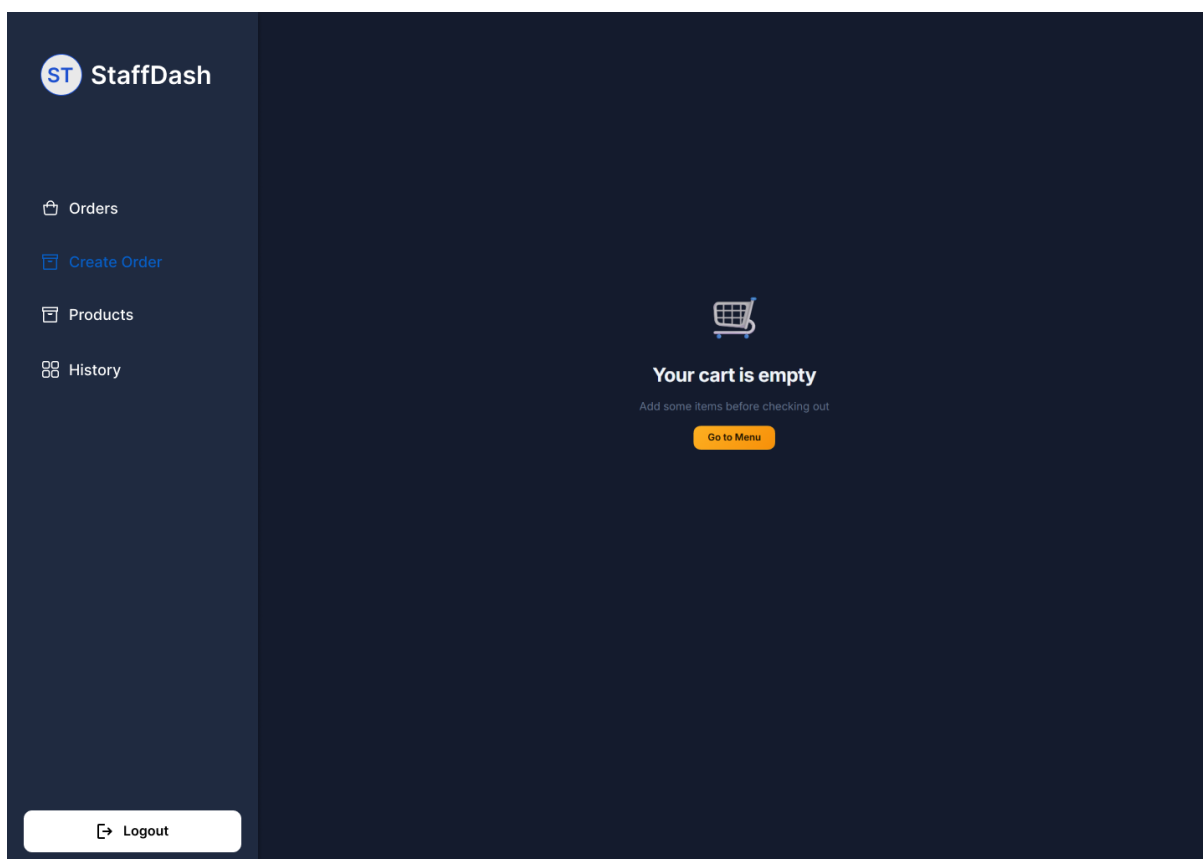
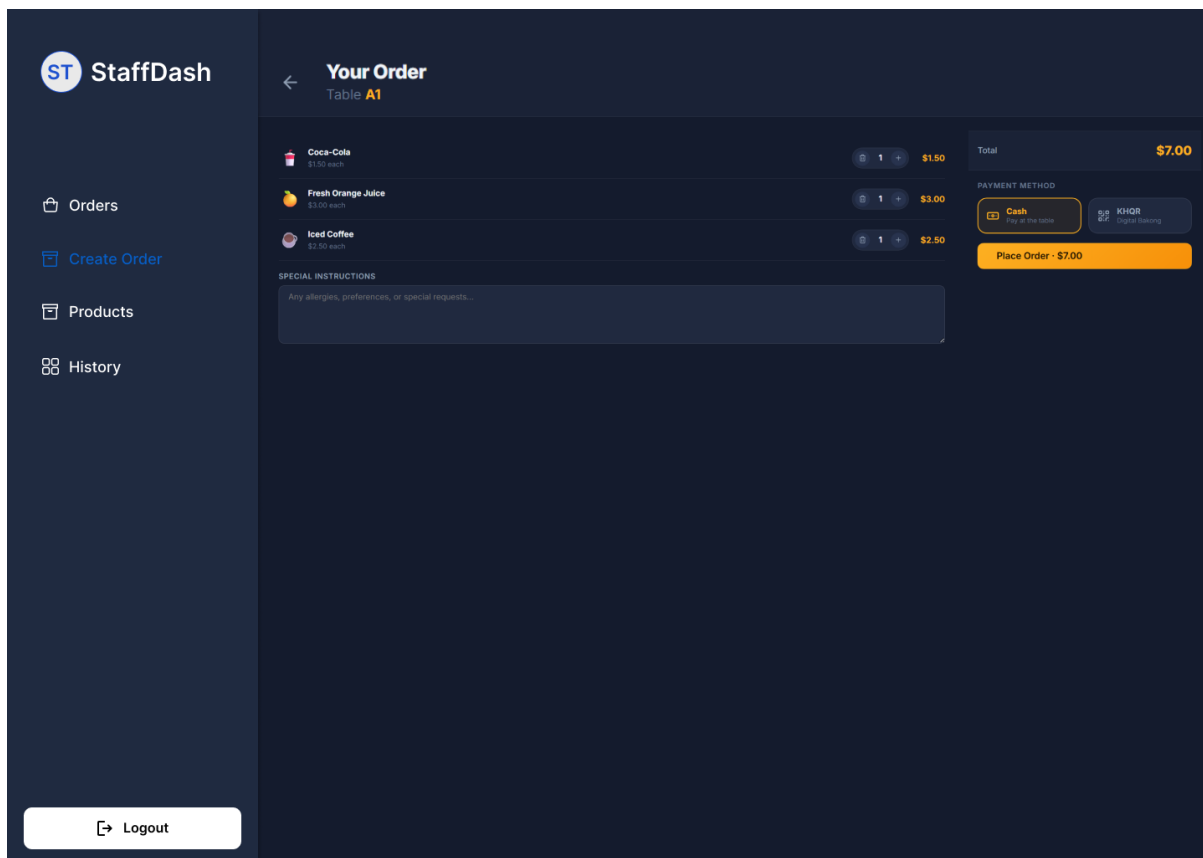
\$3.75

+

1

\$1.50

>





ST StaffDash

Orders

Create Order

Products

History

Logout

Menu Management

13 items — manage availability & details

Q Search products...

All

Drinks

Food

Desserts

Snacks

| ITEM                                                                                             | CATEGORY | PRICE  | AVAILABLE   | ACTIONS                 |
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| <div><div>Chicken Fried Rice</div><div>Wok-fried rice with egg &amp; vegetables</div></div>      | FOOD     | \$4.50 | <div></div> | <div></div> <div></div> |
| <div><div>Grilled Salmon</div><div>Fresh salmon fillet with steamed vegs</div></div>             | FOOD     | \$9.00 | <div></div> | <div></div> <div></div> |
| <div><div>Pad Thai Noodles</div><div>Stir-fried rice noodles, shrimp &amp; peanut...</div></div> | FOOD     | \$5.00 | <div></div> | <div></div> <div></div> |
| <div><div>Mango Sticky Rice</div><div>Classic Thai dessert with sweet mango</div></div>          | DESSERTS | \$3.50 | <div></div> | <div></div> <div></div> |

ST StaffDash

Orders

Create Order

Products

History

Logout

Order History

View recent transactions and reprint receipts

5

ORDERS

\$20.00

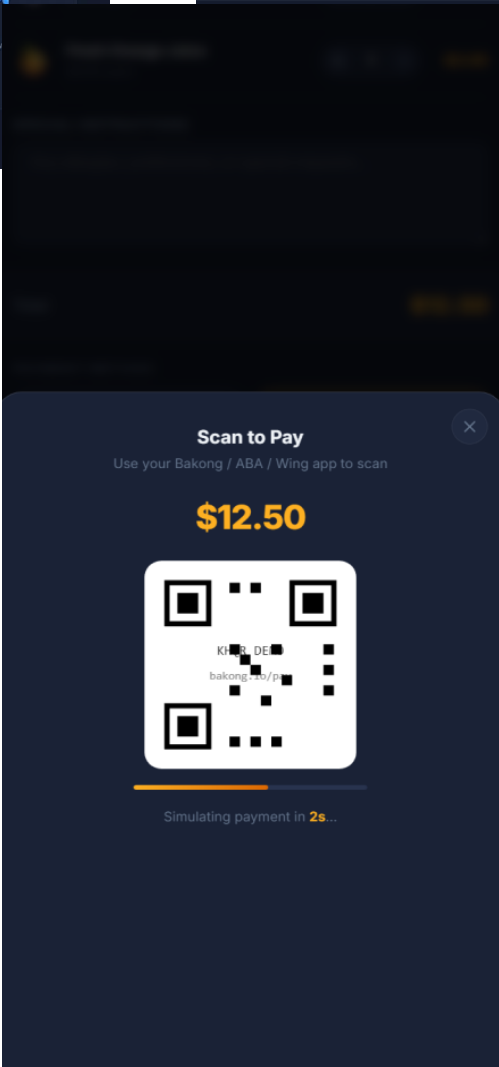
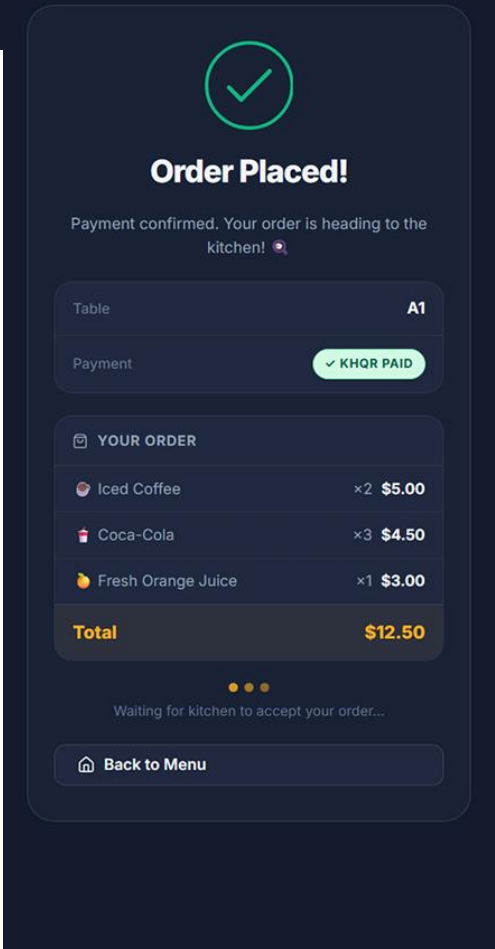
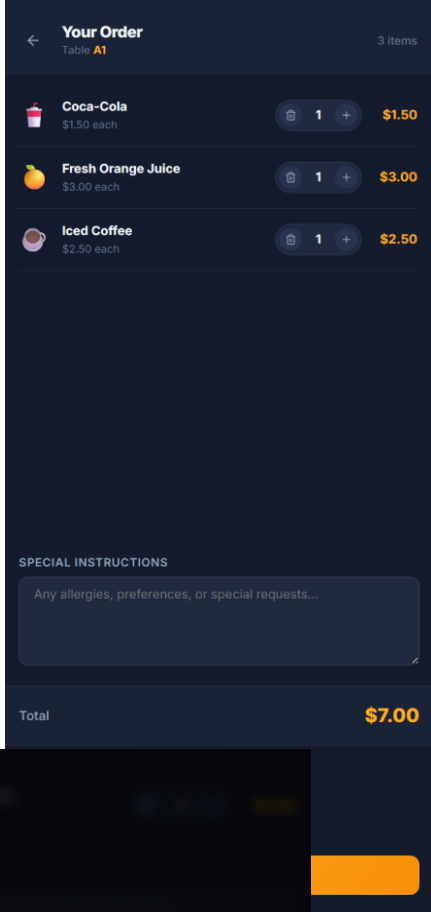
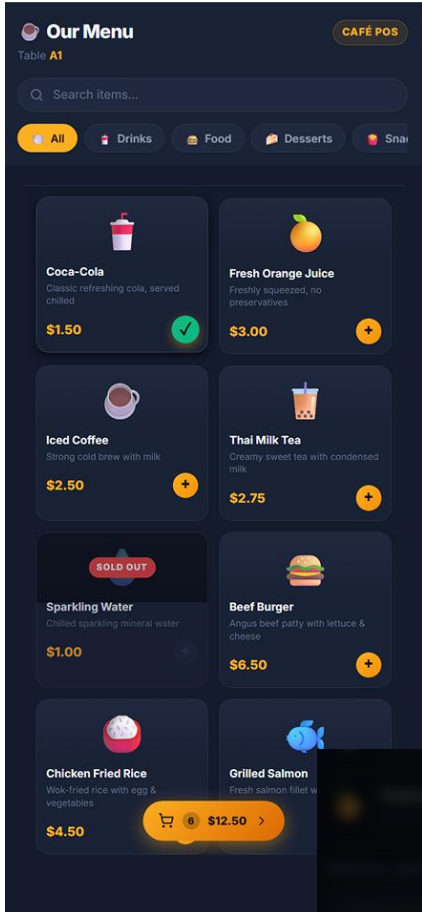
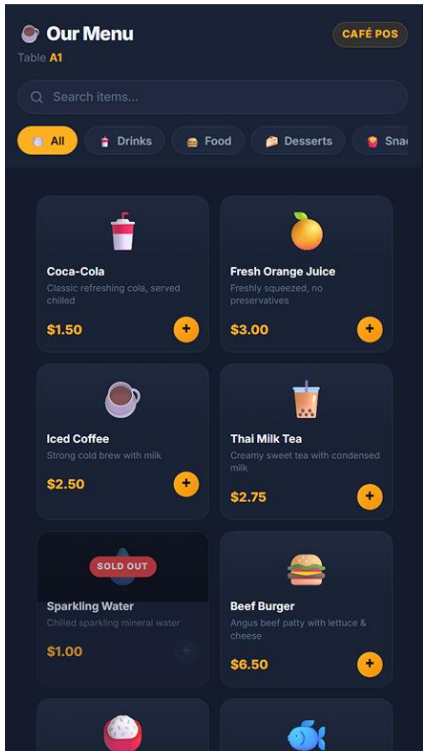
PAID REVENUE

Q Search by Order ID or Table...

Today

| ORDER INFO                                                 | ITEMS                                                                                  | STATUS                                        | TOTAL                                  | ACTIONS          |
|------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------|------------------|
| <div>ORD-005</div> <div>10:41 AM</div> <div>TABLE A5</div> | <div>1x Pad Thai Noodles</div> <div>1x Mango Sticky Rice</div> <div>3x Coca-Cola</div> | <div>UNPAID</div> <div>NOT YET ACCEPTED</div> | <div>\$13.00</div> <div>via CASH</div> | <div>Print</div> |
| <div>ORD-001</div> <div>10:38 AM</div> <div>TABLE A1</div> | <div>2x Coca-Cola</div> <div>1x Beef Burger</div> <div>No onions please</div>          | <div>UNPAID</div> <div>NOT YET ACCEPTED</div> | <div>\$9.50</div> <div>via CASH</div>  | <div>Print</div> |
| <div>ORD-002</div> <div>10:31 AM</div> <div>TABLE B3</div> | <div>2x Iced Coffee</div> <div>1x Crispy Spring Rolls</div>                            | <div>PAID</div> <div>ACCEPTED</div>           | <div>\$8.00</div> <div>via KHQR</div>  | <div>Print</div> |
| <div>ORD-003</div> <div>10:03 AM</div> <div>TABLE A3</div> | <div>1x Grilled Salmon</div> <div>1x Fresh Orange Juice</div>                          | <div>PAID</div> <div>COMPLETED</div>          | <div>\$12.00</div> <div>via KHQR</div> | <div>Print</div> |
| <div>ORD-004</div> <div>09:43 AM</div> <div>TABLE B2</div> | <div>2x Chicken Fried Rice</div> <div>2x Thai Milk Tea</div> <div>Extra spicy</div>    | <div>UNPAID</div> <div>COMPLETED</div>        | <div>\$14.50</div> <div>via CASH</div> | <div>Print</div> |

# Customer



# Test Case

## Customer Module Test Cases for POS

| Test Case ID | Test Scenario                             | Test Steps                                                                                      | Test Data                     | Expected Results                                                                          |
|--------------|-------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------|
| TC-OUST-001  | Accessing menu via table-specific QR code | 1. Use a mobile device to scan the generated QR code.2. Open the browser to the redirected URL. | URL for Table A1 +1           | System opens the menu specifically for Table A1.+1                                        |
| TC-OUST-002  | Filtering products by category            | 1. Tap on different category tabs (Drinks, Food, Snacks).                                       | Category: "Drinks"            | Only items matching the selected category are displayed (e.g., Coca-Cola, Iced Coffee).+3 |
| TC-OUST-003  | Searching for a specific product          | 1. Tap the search bar at the top.2. Type a product name.                                        | Input: "Coffee" +1            | System displays "Iced Coffee" and hides unrelated items.+1                                |
| TC-OUST-004  | Adding an item to the cart                | 1. Find a product in the grid.2. Tap the (+) button.                                            | Product: "Coca-Cola" +1       | Item is added to the cart and the counter increases.                                      |
| TC-OUST-005  | Adjusting item quantity in the cart       | 1. Navigate to "Your Order" (Cart).2. Tap (+) or (-) next to an item.                           | Quantity change: 1 to 2       | Total amount is recalculated immediately to reflect the new quantity.+1                   |
| TC-OUST-006  | Adding special instructions to an order   | 1. Go to the Cart page.2. Tap the "Special Instructions" field and enter a request.             | Text: "No onions please" +2   | The instruction is saved and visible in the order summary.+1                              |
| TC-OUST-007  | Selecting KHQR payment method             | 1. Review cart details.2. Select "KHQR" under Payment Method.3. Tap "Pay with KHQR".            | Selection: "KHQR"             | System generates and displays a dynamic QR code for payment.                              |
| TC-OUST-008  | Successful payment confirmation           | 1. Simulate a successful scan and payment in the app.                                           | Amount: \$12.50               | "Order Placed!" message appears and status changes to "PAID" +1                           |
| TC-OUST-009  | Checking order status after payment       | 1. View the "Order Placed" confirmation screen.                                                 | Current Status                | System displays "Waiting for kitchen to accept your order".                               |
| TC-OUST-010  | Attempting to order a "Sold Out" item     | 1. Locate an item marked as "SOLD OUT".2. Try to tap the (+) button.                            | Product: "Sparkling Water" +1 | The (+) button is disabled or hidden, and the item cannot be added to the cart.           |

| Test Case ID | Test Scenario                             | Test Steps                                                                                 | Test Data                        | Expected Results                                                                    |
|--------------|-------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------|
| TC-ADM-001   | Reviewing total sales analytics           | 1. Navigate to "Dashboard".2. View Revenue and Daily Revenue charts.                       | Dashboard Metrics                | System displays total revenue (e.g., \$1186) and Cash vs. KHQR splits .             |
| TC-ADM-002   | Creating a new menu product               | 1. Navigate to "Products".2. Click "+ Add Item".3. Fill in name, price, and category.      | Name: "Green Tea", Price: \$2.00 | Product is successfully saved and appears in the menu list.                         |
| TC-ADM-003   | Modifying an existing product price       | 1. Go to "Products".2. Click the "Edit" icon next to an item.3. Change the price and save. | Coca-Cola: \$1.50 to \$1.75      | The updated price reflects immediately in both Staff and Customer menus.            |
| TC-ADM-004   | Generating a table-specific QR code       | 1. Navigate to "Table Managements".2. Enter a table number.3. Click "Generator QR".        | Table ID: "A7"                   | System generates a unique URL and QR code image for Table A7 .                      |
| TC-ADM-005   | Identifying 'At Risk' orders in Audit Log | 1. Navigate to "Audit Log".2. Review the list of flagged orders.                           | Audit Criteria                   | System lists orders marked "Served" but still "Unpaid" (e.g., ORD-004 for \$14.50). |
| TC-ADM-006   | Searching extended transaction history    | 1. Navigate to "History".2. Use the search bar for a specific Order ID.                    | Input: "ORD-001"                 | System retrieves the specific historical record regardless of the date .            |
| TC-ADM-007   | Filtering history by date range           | 1. Go to "History" page.2. Select the "Last 7 Days" filter.                                | Filter: "Last 7 Days"            | The table updates to show only transactions from the past week.                     |
| TC-ADM-008   | Deleting an obsolete product              | 1. Navigate to "Products".2. Click the "Delete" (trash) icon.3. Confirm the deletion.      | Item: "Sparkling Water"          | Product is permanently removed from the database and menu.                          |
| TC-ADM-009   | Managing menu categories                  | 1. Navigate to "Categories".2. Add or Edit a category name.                                | Category: "Seasonal"             | The new category becomes available for assigning products.                          |

| Test Case ID | Test Scenario                              | Test Steps                                                                                         | Test Data               | Expected Results                                                                                |
|--------------|--------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------|
| TC-STAFF-001 | Accepting a pending customer order         | 1. Open "Live Dashboard".2. Locate a "Pending" order.3. Click the "Accept" button.                 | Order: ORD-005          | Order moves from "Pending" to "Accepted" status .                                               |
| TC-STAFF-002 | Moving an order to the kitchen             | 1. Locate an "Accepted" order.2. Click "In Kitchen" button.+2                                      | Status: Accepted        | Order status updates to "In Kitchen" for tracking.+2                                            |
| TC-STAFF-003 | Marking an order as "Served"               | 1. Find an order in the "In Kitchen" list.2. Click the "Served" button.+2                          | Order: ORD-001          | Order moves to the "Served" category and is logged for potential audit if unpaid.+2             |
| TC-STAFF-004 | Creating a manual order (Walk-in)          | 1. Click "Create Order".+22. Select Table ID.3. Select items from the grid.4. Click "Place Order". | Table: B3, 2x Latte     | System creates a new order record directly in the dashboard.+3                                  |
| TC-STAFF-005 | Toggling product availability (Sold Out)   | 1. Navigate to "Products".2. Locate "Sparkling Water".3. Switch the "Available" toggle to OFF.     | Item: Sparkling Water   | The item is instantly marked as "SOLD OUT" on the Customer Menu.+1                              |
| TC-STAFF-006 | Searching for a product in menu management | 1. Go to "Products" page.2. Type item name in the search bar.                                      | Input: "Burger"         | Only items containing "Burger" are displayed in the list.                                       |
| TC-STAFF-007 | Viewing limited order history              | 1. Navigate to "History".+12. Check the date filter options.                                       | Filter: Today/Yesterday | Only orders from the current and previous day are visible to the staff.                         |
| TC-STAFF-008 | Reprinting a customer receipt              | 1. Open "History" page.2. Find a specific completed order.3. Click the "Print" button.             | Order: ORD-003          | System triggers the print dialogue for the selected order receipt.                              |
| TC-STAFF-009 | Verifying restricted staff permissions     | 1. Go to the "Products" page.2. Look for "Edit" or "Delete" icons next to items.                   | View: Product List      | Staff can only see the availability toggle; "Edit" and "Delete" actions are hidden or disabled. |

# Test Case

## Staff Module Developer Tasks

| Page           | Functionality               | Task     | SubTask                                                            | Duration | Assign By    | Start Date | End Date |
|----------------|-----------------------------|----------|--------------------------------------------------------------------|----------|--------------|------------|----------|
| Live Dashboard | Real-time Order Management  | Frontend | Build order card component with "Accept" and "Served" buttons.+3   | 25min    | Cheng Lyheng |            |          |
|                |                             |          | Render dashboard grid for Pending, In Kitchen, and Served status . | 20min    |              |            |          |
|                |                             |          | Implement WebSocket listener to display new orders instantly.      | 30min    |              |            |          |
|                |                             | Backend  | Create PATCH /api/orders/:id/status endpoint to update progress.   | 15min    | Cheng Lyheng |            |          |
|                |                             |          | Set up Socket.io event emitter for new order notifications.        | 25min    |              |            |          |
| Create Order   | Manual POS Entry +1         | Frontend | Design table selection dropdown for walk-in orders.                | 15min    | Vann Sak     |            |          |
|                |                             |          | Build searchable product grid for staff selection +1               | 30min    |              |            |          |
|                |                             |          | Create "Place Order" logic for manual Cash/KHOR entries.           | 20min    |              |            |          |
|                |                             | Backend  | Create POST /api/orders/manual with staff-assigned Table ID.       | 20min    | Vann Sak     |            |          |
|                |                             |          | Implement logic to mark manual orders as "PAID" or "UNPAID".       | 15min    |              |            |          |
| Products       | Menu Availability Toggle +1 | Frontend | Implement binary toggle switch for "Available" status.             | 15min    | Cheng Lyheng |            |          |
|                |                             |          | Hide "Edit/Delete" actions based on Staff role permissions.        | 10min    |              |            |          |
|                |                             |          |                                                                    |          |              |            |          |
|                |                             | Backend  | Create PATCH /api/products/:id/availability for the toggle.        | 15min    | Cheng Lyheng |            |          |
|                |                             |          | Validate that ONLY the is_available field can be edited by Staff.  | 10min    |              |            |          |
| History        | Recent Transaction Support  | Frontend | Create "Today/Yesterday" filter logic for the transaction list.    | 15min    | Chey Kimrith |            |          |
|                |                             |          | Add "Print Receipt" trigger for specific orders.                   | 15min    |              |            |          |
|                |                             |          |                                                                    |          |              |            |          |
|                |                             | Backend  | Create GET /api/history/staff with a 48-hour date limit.           | 20min    | Chey Kimrith |            |          |
|                |                             |          | Implement search query logic for Order ID or Table Number.         | 15min    |              |            |          |

## Customer Module Developer Tasks

| Page                | Functionality               | Task     | SubTask                                                      | Duration | Assign By     | Start Date | End Date |
|---------------------|-----------------------------|----------|--------------------------------------------------------------|----------|---------------|------------|----------|
| Digital Menu (Home) | Product Browsing & Search   | Frontend | Create Category Tabs UI (All, Drinks, Food, etc)             | 20min    | Cheng Lyheng  |            |          |
|                     |                             |          | Design Product Grid with "Sold Out" overlay logic            | 30min    |               |            |          |
|                     |                             |          | Implement Search Bar real-time filter logic                  | 15min    |               |            |          |
|                     |                             | Backend  | Create GET /api/products with category filtering             | 15min    | Cheng Lyheng  |            |          |
|                     |                             |          | Write Prisma query for is_available check                    | 10min    |               |            |          |
|                     |                             |          |                                                              |          |               |            |          |
| Your Order (Cart)   | Cart & Special Instructions | Frontend | Build Cart Service to manage local state (Qty +/-)           | 25min    | Chey Kimrith  |            |          |
|                     |                             |          | Create Special Instructions text area                        | 10min    |               |            |          |
|                     |                             |          | Implement Real-time Total Calculation logic                  | 10min    |               |            |          |
|                     |                             | Backend  | Create POST /api/orders to save items and instructions       | 20min    | Chey Kimrith  |            |          |
|                     |                             |          | Validate order items against current stock/availability      | 15min    |               |            |          |
|                     |                             |          |                                                              |          |               |            |          |
| Scan to Pay         | KHQR Payment Integration    | Frontend | Design KHQR Display component with Simulation timer          | 20min    | Sak/rith/heng |            |          |
|                     |                             |          | Implement polling for payment status confirmation            | 15min    |               |            |          |
|                     |                             |          |                                                              |          |               |            |          |
|                     |                             | Backend  | Integrate Bakong/KHOR SDK for dynamic QR generation          | 40min    | Sak/rith/heng |            |          |
|                     |                             |          | Create Webhook endpoint for payment success callback         | 25min    |               |            |          |
| Order Status        | Real-time Order Tracking    | Frontend | Create Order Status stepper (Placed -> In Kitchen -> Served) | 20min    | Vann Sak      |            |          |
|                     |                             |          | Implement WebSocket listener for live status updates         | 30min    |               |            |          |
|                     |                             |          |                                                              |          |               |            |          |
|                     |                             | Backend  | Set up WebSocket gateway for status change notifications     | 35min    | Vann Sak      |            |          |
|                     |                             |          | Update order_status in DB on Staff/Admin action              | 10min    |               |            |          |

## Admin Module Developer Tasks

| Page             | Functionality                   | Task     | SubTask                                                                     | Duration | Assign By    | Start Date | End Date |
|------------------|---------------------------------|----------|-----------------------------------------------------------------------------|----------|--------------|------------|----------|
| Sales Analytics  | Business Performance Monitoring | Frontend | Integrate Chart.js/ApexCharts for Daily Revenue (Cash vs KHQR).             | 40min    | Cheng Lyheng |            |          |
|                  |                                 |          | Build Revenue Split pie chart component.                                    | 20min    |              |            |          |
|                  |                                 |          | Design "Top Products" data list with volume tracking.                       | 15min    |              |            |          |
|                  |                                 | Backend  | Create aggregate API for total revenue and order counts.                    | 30min    | Cheng Lyheng |            |          |
|                  |                                 |          | Implement grouping logic by payment method and product frequency.           | 25min    |              |            |          |
| Menu Management  | Full Inventory Control          | Frontend | Build CRUD models for Add, Edit, and Delete products.                       | 45min    | Vann Sak     |            |          |
|                  |                                 |          | Implement form validation for prices and category assignments.              | 20min    |              |            |          |
|                  |                                 | Backend  | Create full CRUD endpoints (POST, PUT, DELETE) for Products and Categories. | 40min    | Vann Sak     |            |          |
|                  |                                 |          | Implement image upload handling for menu item thumbnails.                   | 30min    |              |            |          |
| Table Setup      | QR System Configuration         | Frontend | Build input form for Table Numbers (e.g., A1, B5).                          | 15min    | Cheng Lyheng |            |          |
|                  |                                 |          | Design "Generator QR" trigger and image download logic.                     | 20min    |              |            |          |
|                  |                                 |          | Create logic to generate unique, encrypted URLs for each table ID.          | 25min    | Cheng Lyheng |            |          |
|                  |                                 | Backend  | Implement QR code generation library (e.g., qrcode npm).                    | 20min    |              |            |          |
| Audit Log        | Financial Loss Prevention       | Frontend | Build "At Risk" list view for served-but-unpaid orders.                     | 25min    | Chey Kimrith |            |          |
|                  |                                 |          | Display "Total at Risk" summary banner.                                     | 10min    |              |            |          |
|                  |                                 |          | Create query to flag orders where status: SERVED and payment: UNPAID.       | 20min    | Chey Kimrith |            |          |
|                  |                                 | Backend  | Implement risk amount summation logic.                                      | 15min    |              |            |          |
| All Transactions | Extended Audit Trail            | Frontend | Build advanced filtering UI (Date Range, Payment Status).                   | 30min    | Chey Kimrith |            |          |
|                  |                                 |          | Implement global search by Order ID or Table.                               | 15min    |              |            |          |
|                  |                                 |          | Create high-performance GET endpoint with pagination for historical data.   | 30min    | Chey Kimrith |            |          |
|                  |                                 | Backend  | Optimize search indexing for order_id and table_no.                         | 20min    |              |            |          |